

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C++, Python, TypeScript/JavaScript, C, SQL, Java

Test & Automation: Software-in-the-loop, automated regression, scripted replay, parameter sweeps, pytest, GitHub Actions

Software & APIs: FastAPI, Flask, WebSockets, REST/JSON, SQLite, Docker Compose, structured logging

Embedded & Data: CAN, TTC 60 VCU, STM32H7, PX4, GDB, pandas, NumPy, CSV/JSON telemetry

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead **12 engineers** across **10+ embedded, test, and telemetry projects**; define code standards and build, release, and test-integration plans for control-unit firmware
- Developing the custom-inverter VCU path for a **four-motor** EV, mapping calibrated pedal input to bounded CAN current commands on a **10 ms** loop with fail-safe precharge and ready-to-drive gating
- Reworked VCU integration for a custom BMS, decoding **46 CAN message IDs** across **96 cells**, **96 thermistors**, and **8 modules**, with a **1 s** heartbeat timeout for communication loss

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated flight-control changes in C on PX4, using structured telemetry logs and repeatable flight tests to reproduce issues and evaluate revisions
- Integrated and tuned visual-inertial odometry for onboard state estimation, comparing configuration changes against logged position-hold behavior

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Designed a lap-adaptive energy controller that derived its power setpoint from cumulative energy versus budget and held measured power within **0.01%** of the commanded setpoint
- Developed fixed-point PI firmware with saturation, anti-windup, and slew limiting to maximize motor output while enforcing an **80 kW** limit
- Built Python telemetry parsers and a software-in-the-loop replay harness for recorded CSV logs, controller parameter sweeps, and regression review

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Tuned the PI torque controller used for power limiting to within **2%** error using logged step responses, smoothing torque transitions to preserve drivability

PROJECT EXPERIENCE

SR-Wireless-CAN — Firmware Deployment & Test Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, FastAPI, WebSockets, React, SQLite, Docker Compose, CAN

- Led backend development for a vehicle test platform supporting **3 capture modes** — simulated, recorded-trace replay, and live CAN — plus **3 signal-monitoring views**
- Reverse-engineered and implemented an **8-stage** CAN firmware-deployment workflow from vendor traces, with CRC verification, error logging, and persisted flash history
- Containerized the backend, frontend, and database for one-command setup and documented protocol behavior, configuration, and recovery steps for repeatable team use